Correct Answer Distribution Across Two Paths

Thesis-formatted result · accuracy-first tail-leakage gate τ = 0.002305

Thesis Table 5.8

Table 5.8: Correctness overlap between the direct and decomposition candidates before final routing at the accuracy-first operating point ($\tau = 0.002305$). Counts are out of 1,000 questions per dataset, and the four columns are mutually exclusive.

Dataset	Both paths correct	Only direct correct	Only decomposition correct	Neither path correct
HotpotQA	225	78	135	562
2Wiki	144	36	353	467
MuSiQue	75	27	192	706
PopQA	346	32	194	428

Decomposition-only wins exceed direct-only wins on every dataset.

Verifier Path Selection

Thesis-formatted result · accuracy-first tail-leakage gate τ = 0.002305

Thesis Table 5.7

Table 5.7: Average plan length and verifier choices at the accuracy-first operating point ($\tau = 0.002305$). Selection counts and percentages use all 1,000 questions per dataset.

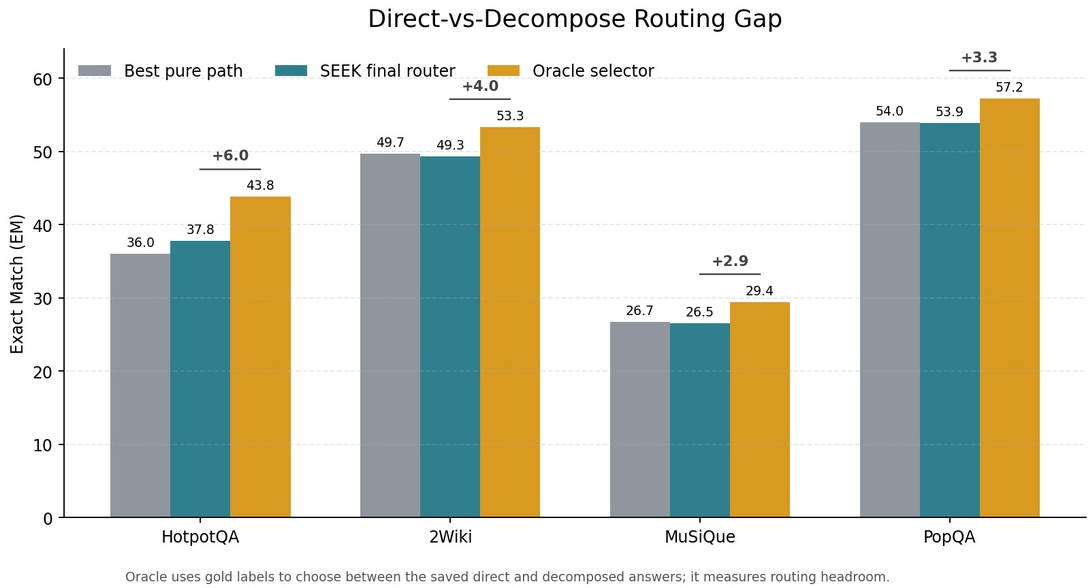
Dataset	Mean sub-questions	N	Direct chosen n (%)	Decomposition chosen n (%)
HotpotQA	2.26	1,000	171 (17.1%)	829 (82.9%)
2Wiki	2.47	1,000	25 (2.5%)	975 (97.5%)
MuSiQue	2.03	1,000	121 (12.1%)	879 (87.9%)
PopQA	1.00	1,000	10 (1.0%)	990 (99.0%)
Pooled	1.94	4,000	327 (8.2%)	3,673 (91.8%)

The verifier chooses decomposition 91.8% of the time while retaining a direct fallback.

Direct-vs-Decompose Oracle Gap

Thesis-formatted result · accuracy-first tail-leakage gate τ = 0.002305

Thesis Figure 5.5



Oracle uses gold labels to choose between the saved direct and decomposed answers; it measures routing headroom.

A perfect path selector still offers +4.05 EM on average; HotpotQA has the largest gap (+6.0).